

KOVVURI AKSHA

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Education

Manipal Institute of Technology, Karnataka

Aug 2023 – July 2027

B.Tech in Computer Science and Engineering (CGPA: 7.87/10)

Lakshya School

2021 – 2023

Class 12 CBSE Board (Percentage: 91.4%)

Projects

Rooftop Solar Potential Mapping | *Data Science & Forecasting*

GitHub

- **Analyzed 8 years** (2015-2023) of satellite-derived solar irradiation data from NASA POWER and urban building footprints from **OpenStreetMap** across **5 major NCR cities**.
- **Cleaned, imputed, and standardized** large-scale multi-source geospatial datasets covering 4.8M+ buildings.
- Built baseline and non-linear regression models (**Linear Regression, Random Forest**) to forecast capacity through 2030.
- Delivered forecasting insights to support urban energy planning and policy decision-making with **fine-tuned Random Forest model** ($R^2 = 0.984$, **RMSE = 274 MW**).
- Conducted error analysis and cross-validation to assess model robustness and limitations.
- Performed **feature importance analysis** and showcased **visualizations** to highlight spatial trends and policy-relevant insights.

COVID-19 & AQI Analysis | *Exploratory Data Science*

GitHub

- Performed **exploratory data analysis** on COVID-19 recovery rates and air quality (AQI) data across 5 major Indian cities.
- Handled missing and inconsistent multi-source time series data using **statistical imputation** and **normalization techniques**.
- Visualized temporal trends and correlations using **Matplotlib** and **D3.js** to communicate insights effectively.

Experience

Amaravathi Solutions | *Software Intern*

Dec 2025 – Jan 2026

- Developed and maintained **RESTful APIs** using **Node.js** and **Express** for core backend services to handle traffic of 10-50 requests/sec.
- Optimized database schemas and indexes to improve performance of aggregation-heavy analytical queries to reduce response time below 200 ms.
- Implemented **authentication and authorization** mechanisms to secure backend endpoints.
- Debugged and resolved backend issues, improving overall **system stability and performance**.

Technical Skills

Languages	Python, SQL
Data Science	Pandas, NumPy, Exploratory Data Analysis (EDA), Time Series Analysis, Statistical Modeling
Machine Learning	Scikit-learn, PyTorch
Statistics	Hypothesis Testing, Model Validation, Error Analysis
Visualization	Matplotlib, Seaborn, D3.js
Databases & Tools	MongoDB, Git, Postman

Relevant Coursework

Data Structures and Algorithms | Operating Systems | C Programming | Statistical Methods in AI | Probability and Random Processes | Linear Algebra | Computer Networks | Object Oriented Programming

Extracurricular

- **NTSE Scholar** (Stages 1 & 2)
- **Amazon ML Hackathon – FusionPrice** (Participant)
 - Solved a price prediction problem using only product images as input, applying machine learning models with brand, quantity, and location attributes as major contributors.
 - Used key contributing features to train regression models, achieving over 90% accuracy in predicting optimal product pricing for a real-world marketplace scenario.
- **Volunteering**
 - In collaboration with IEEE and the NSS program, taught students the essential fundamentals of Emergency Response, Safety Protocols and Disaster Preparedness through interactive sessions.
 - Provided community outreach education demonstrating leadership, communication, and social responsibility.